

**Benchmarking Gaze Biomarkers: A Cross-Dataset Evaluation of
Kinematic, Physiological, and Statistical Features for ADHD
Detection**

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On my honor as a University Student, I have neither given nor received unauthorized aid on this assignment as defined by the Honor Guidelines for Thesis-Related Assignments

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ABSTRACT

ADHD diagnosis remains limited by subjective assessment, motivating interest in eye-tracking as an objective digital biomarker. Gaze behavior captures spatial attention, while pupil dynamics reflect arousal, effort, and control processes. Prior work shows that both signals contain ADHD-related information, but most methods study them separately or fuse hand-crafted features. This project evaluates gaze-only and pupil-only baselines and introduces FusionADHDNet, an end-to-end multimodal model that jointly learns from pupil signals, gaze trajectories, missingness masks, and task metadata. On the FigShare ADHD eye-tracking dataset, FusionADHDNet improved trial-level AUC and balanced accuracy over both reproduced baselines. However, low specificity under the selected threshold shows that better calibration, stronger class balance, and richer subject-level data are needed before clinical use.

INTRODUCTION

ADHD diagnosis relies heavily on clinical judgment, questionnaires, and behavioral observation. These tools are useful but can be affected by rater bias, context, and symptom variability. Eye-tracking offers a

noninvasive way to measure attention and physiological regulation more directly.

Two signals are central to this work: gaze and pupil dynamics. Gaze captures where and how attention moves through space over time. Pupil size reflects arousal, cognitive effort, and internal control. Prior studies show that both are useful for ADHD classification, but most models treat them separately. Gaze models often depend on scanpaths and stimulus saliency, while pupil models focus on temporal pupil traces and missing-data handling.

This project addresses that gap by testing whether a multimodal model can learn stronger ADHD-related representations than either modality alone. We compare a Deng-inspired gaze baseline, a Choi-inspired pupil baseline, and FusionADHDNet, a proposed model that combines gaze, pupil, masks, and task metadata in one trainable architecture. The goal is not to replace diagnosis, but to test whether multimodal eye-tracking can support more reliable objective screening.

RELATED WORKS

Prior ADHD eye-tracking work falls into three areas: gaze modeling, pupil modeling, and multimodal fusion.

Gaze-based methods model how subjects explore visual scenes through fixations, saccades, and scanpaths. Deng et al. proposed an end-to-end ADHD classifier using fixation coordinates, fixation duration, and stimulus-derived saliency during natural viewing. This showed that gaze is most meaningful when interpreted with scene context. However, performance varied across videos, suggesting that gaze-only models may depend heavily on stimulus conditions.

Pupil-based methods use pupil diameter as a physiological marker of arousal and cognitive effort. Earlier work used engineered pupil features, while Choi et al. introduced a deep model that jointly imputes missing pupil values and predicts ADHD. This is important because pupil recordings often contain gaps from blinking, motion artifacts, and tracking noise. Treating missingness as part of the learning problem makes pupil modeling more realistic for raw eye-tracking data.

Multimodal work suggests that gaze and pupil signals are complementary. Ueno et al. combined eye-movement complexity features with mean pupil diameter and classified ADHD using logistic regression. Their results support multimodal biomarkers, but the fusion occurs after hand-crafted feature extraction. Our work moves fusion inside the network by learning directly from raw temporal gaze and pupil representations.

PROJECT DESIGN

This project compares two reproduced baselines with a proposed multimodal model.

The final dataset contained 10,720 trial-level examples from 50 subjects, including 3,520 ADHD trials and 7,200 non-ADHD trials. Each trial included subject ID, label, trial

number, cognitive load, distractor condition, reaction time, pupil time series, and gaze coordinates. Pupil sequences were length 8,000. Gaze sequences were variable length, averaging 7,223 samples.

All signals were standardized before training. Pupil traces were interpolated over missing regions, paired with an observation mask, z-scored per trial, and resampled to 512 time steps. This produced two pupil channels: signal and mask. Gaze coordinates were interpolated, z-scored, resampled to 512 time steps, and augmented with velocity channels. This produced five gaze channels: x, y, dx, dy, and mask.

The Deng-inspired baseline used a temporal convolutional encoder over gaze-derived channels. Because our dataset did not include natural-viewing saliency maps, this baseline tested whether gaze dynamics alone transferred to the FigShare task setting.

The Choi-inspired baseline used pupil signal and mask channels with a temporal convolutional encoder, reconstruction head, and classification head. Its loss combined binary classification with masked pupil reconstruction, encouraging the model to preserve useful pupil dynamics.

FusionADHDNet combines both modalities. It has a pupil encoder, gaze encoder, pupil-dominant cross-attention fusion block, metadata encoder, hybrid gating module, classifier, and auxiliary pupil reconstruction head. The pupil branch receives signal and mask channels. The gaze branch receives x, y, dx, dy, and mask channels. Each branch uses ConvBlock1D stages with convolution, batch normalization, GELU activation, dropout, residual connections, and dilation.

After encoding, pupil and gaze features are projected into a shared 64-dimensional space and aligned to a fusion length of 192. The pupil sequence queries the gaze sequence through cross-attention, allowing pupil dynamics to be updated with spatial-attention context. The fused representation is combined with metadata, including load, distractor condition, reaction time, and signal-quality features. The final classifier outputs ADHD probability. During training, random pupil regions are masked and reconstructed with an auxiliary loss weighted by $\lambda = 0.08$.

RESULTS

We evaluated three models: a Deng-inspired gaze baseline, a Choi-inspired pupil baseline, and FusionADHDNet. All models produced trial-level probabilities, which were also aggregated by subject. Metrics included AUC, balanced accuracy, sensitivity, and specificity because the dataset was imbalanced.

The Deng-inspired gaze baseline achieved 0.49 trial AUC, 0.48 balanced accuracy, 0.59 sensitivity, and 0.37 specificity. This was the weakest result. The likely reason is that the original Deng model benefits from stimulus saliency and natural-viewing context, while this reproduction used gaze trajectories without scene-level information. Gaze position alone was therefore under-specified.

The Choi-inspired pupil baseline achieved 0.62 trial AUC and 0.50 balanced accuracy, with 1.00 sensitivity and 0.00 specificity under the selected threshold. Its reconstruction objective helped it learn a better ranking than the gaze baseline, but its binary decisions were poorly balanced. This suggests that pupil-only modeling captured some physiological signal but did not produce a reliable class boundary.

FusionADHDNet achieved the strongest overall results: 0.75 trial AUC, 0.59 balanced accuracy, 0.92 sensitivity, and 0.26 specificity. This improved over the Deng baseline by about 0.26 AUC and 0.11 balanced accuracy, and over the Choi baseline by about 0.13 AUC and 0.09 balanced accuracy. The main gain was AUC, meaning the fusion model learned a stronger continuous ranking of trials than either unimodal baseline.

These results support the hypothesis that gaze and pupil signals contain complementary information. Pupil dynamics may encode arousal and effort, while gaze features provide spatial-attention context. However, the low specificity is a major limitation. Under the selected threshold, FusionADHDNet favored sensitivity and struggled to reject non-ADHD cases. For screening, high sensitivity is useful, but low specificity would create too many false positives.

The specificity problem likely reflects class imbalance, limited subject count, variable gaze quality, and repeated trial-level labels within subjects. Although the dataset contains many trials, it only includes 50 subjects, so trial-level performance may overstate generalization. A stronger dataset would include more subjects, a more balanced ADHD/control split, synchronized stimulus information, cleaner gaze recordings, and multiple tasks per subject.

Overall, FusionADHDNet improved representation learning but is not clinically reliable yet. The higher AUC shows promise, while the low specificity shows the need for better calibration, thresholding, class-balanced losses, and independent subject-level validation.

CONCLUSION

This project evaluated ADHD classification from eye-tracking signals using gaze-only, pupil-only, and multimodal models. The unimodal baselines captured useful but incomplete information: gaze modeled spatial attention, while pupil modeled physiological dynamics and missingness. FusionADHDNet achieved the best trial-level AUC and balanced accuracy by combining pupil signals, gaze trajectories, masks, and metadata in one end-to-end architecture.

The results support multimodal eye-tracking as a promising direction for objective ADHD screening. However, the model's low specificity shows that better AUC does not guarantee clinical reliability. Future systems need stronger calibration, larger and more balanced datasets, and stricter subject-independent validation before they can support real diagnostic decisions.

FUTURE WORK

Future work should focus on four improvements. First, threshold calibration should be improved using validation-based thresholding, class-balanced losses, focal loss, and calibration curves. Second, the model should be tested on larger datasets with more subjects and a more balanced ADHD/control split. Third, future datasets should include synchronized gaze, pupil, timestamps, and stimulus information so scanpaths can be interpreted in context. Fourth, ablation studies should test the contribution of each component, including pupil reconstruction, gaze velocity, masks, metadata, cross-attention, and hybrid gating.

These steps would clarify whether FusionADHDNet's improvement comes from true multimodal learning and whether it generalizes beyond this dataset.

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